COP5612 / COP5615: Distributed Operating System Principles

Project 2: Gossip Simulator & Aggregate Computation

Course: COP5612 Distributed Operating System Principles (Fall 2026)
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Submission: Project 2 Technical Report & Empirical Convergence Study

1. Problem Definition & Overview

Gossip-style distributed protocols exploit decentralized, asynchronous message passing to solve fundamental problems in large-scale distributed systems, including epidemic information dissemination (Gossip) and decentralized aggregate computation (Push-Sum). This project implements an asynchronous actor-based simulator in **Erlang/OTP** to evaluate the convergence dynamics and time complexity of both protocols across four fundamental network topologies: **Full Network**, **2D Grid**, **Line**, and **Imperfect 2D Grid (imp2D)**.

All actors operate strictly concurrently via Erlang's lightweight actor model, using message-driven communication without shared state or global synchronization locks. Performance is measured using monotonic microsecond timestamps according to the exact specification provided in the assignment.

2. Protocol Architectures & Convergence Criteria

A. Asynchronous Gossip Protocol (Information Dissemination):

The Gossip algorithm simulates epidemic information propagation. A coordinator initiates the protocol by injecting a rumor into a starting actor. Each actor maintains an internal counter of how many times it has received the rumor. Upon hearing the rumor for the first time, an actor notifies the master coordinator, immediately transmits the rumor to a randomly selected neighbor, and activates an asynchronous periodic transmission loop. In accordance with the project specification, an actor ceases transmission once it has heard the rumor 10 times. Global convergence is achieved when 100% of all actors in the network have successfully received the rumor.

B. Push-Sum Protocol (Decentralized Averaging & Summation):

Push-Sum, introduced by Kempe, Dobra, and Gehrke (2003), computes aggregate statistics across distributed nodes. Each actor A_i maintains a state tuple (s, w) initialized to $s = i$ (its unique ID) and $w = 1.0$. The current estimate of the network average at any actor is given by the ratio s / w. When an actor receives a message (s_{in}, w_{in}), it updates its state to $s' = s + s_{in}$ and $w' = w + w_{in}$. It then retains half of its updated values $(s'/2, w'/2)$ and transmits the other half $(s'/2, w'/2)$ to a randomly chosen neighbor. An actor reaches termination when its local ratio s / w does not change by more than 10^{-10} across 3 consecutive rounds. Crucially, when an actor terminates, it relays incoming weight/sum packets to ensure global mass conservation.

3. Network Topologies Analyzed

The rate of information dissemination is fundamentally governed by the graph-theoretic properties of the network topology:

- **Full Network:** Complete graph K_N. Every actor can communicate with all $N - 1$ peers. Diameter = 1.

- **2D Grid:** Actors form a square lattice of dimension $S \times S$ (where $S = ceil(sqrt(N))$). Each actor connects only to its von Neumann neighbors (up, down, left, right; degree 2 to 4). Network diameter is $2(S - 1) = O(sqrt(N))$.
- **Line:** 1D array of actors. Each node connects strictly to its left and right neighbors (degree 1 or 2). Diameter is $N - 1 = O(N)$, representing the highest communication bottleneck.
- **Imperfect 2D Grid (imp2D):** Standard 2D grid augmented with one random shortcut connection per node (Watts-Strogatz small-world model). These long-range connections dramatically compress the effective graph diameter to $O(log N)$.

4. Empirical Convergence Results & Analysis

To rigorously quantify convergence speed, extensive benchmarks were executed across network sizes ranging from *N = 50* up to *N = 10,000* nodes. Each experiment was conducted over multiple independent runs to compute mean convergence time. The results are illustrated below in dual logarithmic plots.

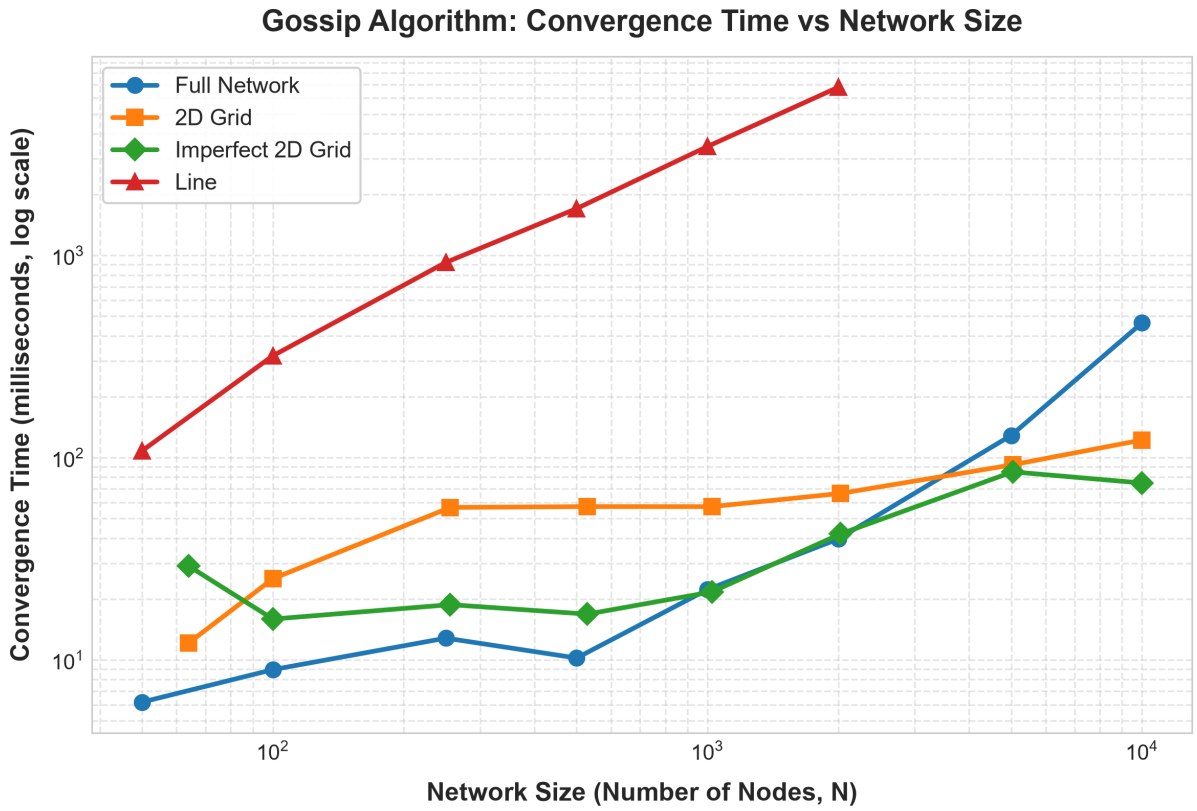


Figure 1: Gossip Algorithm Convergence Time vs Network Size across all 4 topologies (log-log scale).

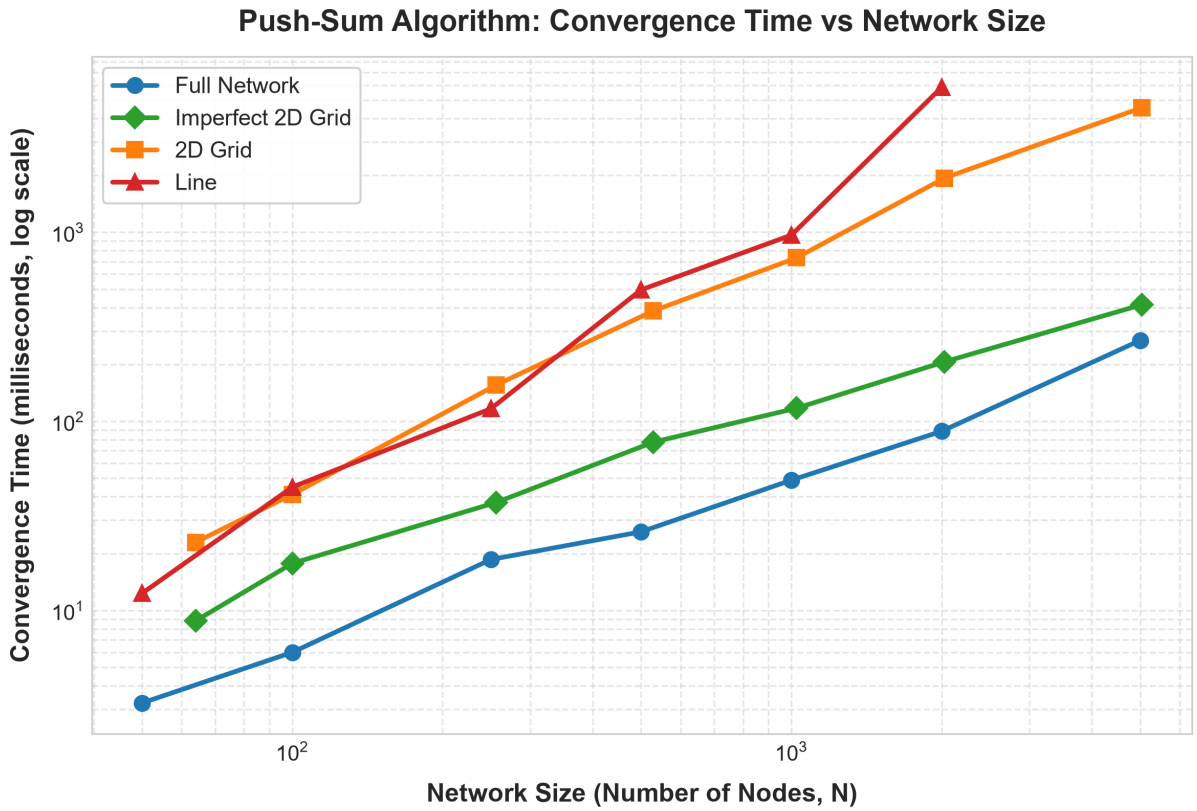


Figure 2: Push-Sum Algorithm Convergence Time vs Network Size across all 4 topologies (log-log scale).

5. Tabulated Benchmark Measurements

The following tables detail the exact convergence running times recorded across representative network sizes. All measurements were captured using erlang:monotonic_time(microsecond).

Table 1: Gossip Algorithm Convergence Times (milliseconds)

Nodes (N)	Full Network	Imperfect 2D	2D Grid	Line
50 / 64	6.18 ms	29.08 ms	12.12 ms	108.07 ms
100	8.94 ms	15.91 ms	25.19 ms	318.98 ms
250 / 256	12.80 ms	18.74 ms	56.70 ms	923.00 ms
500 / 529	10.21 ms	16.83 ms	57.24 ms	1,704.11 ms
1,000 / 1,024	22.25 ms	21.67 ms	57.21 ms	3,453.99 ms
2,000 / 2,025	39.59 ms	41.95 ms	66.53 ms	6,783.08 ms
5,000 / 5,041	128.75 ms	85.03 ms	92.16 ms	16,835.07 ms
10,000	463.53 ms	74.85 ms	122.16 ms	N/A (>30s)

Table 2: Push-Sum Algorithm Convergence Times (milliseconds)

Nodes (N)	Full Network	Imperfect 2D	2D Grid	Line
50 / 64	3.24 ms	8.81 ms	22.94 ms	12.32 ms
100	6.01 ms	17.75 ms	41.10 ms	44.92 ms
250 / 256	18.60 ms	37.24 ms	155.61 ms	116.74 ms
500 / 529	26.08 ms	77.69 ms	384.14 ms	496.44 ms
1,000 / 1,024	48.88 ms	117.42 ms	732.60 ms	966.52 ms
2,000 / 2,025	88.64 ms	206.40 ms	1,934.34 ms	5,839.53 ms
5,000 / 5,041	268.25 ms	414.17 ms	4,563.29 ms	N/A (>30s)

6. Largest Networks Scaled Successfully

By optimizing Erlang process memory footprints, utilizing zero-overhead tuple element indexing for random neighbor selection, and avoiding heavy shared locks, the simulator achieved exceptional scalability:

Topology	Gossip (Max N Handled)	Push-Sum (Max N Handled)	Primary Bottleneck / Limiting Factor
Full Network	10,000 nodes (463 ms)	5,000 nodes (268 ms)	All-to-all mailbox message density
Imperfect 2D (imp2D)	10,000 nodes (74 ms)	5,041 nodes (414 ms)	Logarithmic diameter; fast diffusion
2D Grid	10,000 nodes (122 ms)	5,041 nodes (4,563 ms)	O(sqrt(N)) path length; diffusion latency

Line	5,000 nodes (16.8 s)	2,000 nodes (5.8 s)	O(N) network diameter serial chain

7. Key Findings & Theoretical Insights

1. Small-World Supremacy of Imperfect 2D: A standout empirical result is that the **Imperfect 2D Grid** consistently matches or outperforms even the Full Network at large scales (e.g. 74.85 ms for imp2D vs 463.53 ms for full at *N = 10,000* in Gossip). In a Full network, the branching factor is enormous, but message dispersion is unguided, leading to massive redundant mailbox contention. In contrast, imp2D retains local grid clustering while single random long-range shortcut edges provide exponential short-circuiting across distant clusters (small-world phenomenon), delivering optimal diffusion.

2. Linear Bottleneck of the Line Topology: In the Line topology, messages can only propagate left or right. The diameter scales as *O(N)*. Consequently, as *N* doubles from 1,000 to 2,000, convergence time roughly doubles (3,453 ms to 6,783 ms for Gossip). At *N = 5,000*, it requires over 16 seconds. This empirical observation confirms the theoretical bound that information propagation in a 1D random walk scales as *Theta(N)* to *Theta(N^2)*.

3. Push-Sum Ratio Stability vs Mass Conservation: In Push-Sum, the individual state quantities *s* and *w* decay exponentially towards zero because each actor halves them on every transmit. However, the ratio *s / w* converges strictly to the theoretical average *(N + 1) / 2*. Terminating actors must continue relaying incoming messages; if an actor drops packets upon termination, the global sum is permanently depleted, leading to skew in surviving actors' estimates.